SYNDRI: Context-aware removal of transcript spillover in imaging-based spatial transcriptomics

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Abstract

Imaging-based spatial transcriptomics (ST) measures gene expression at single-cell resolution, but imperfect segmentation often misassigns transcripts between neighboring cells, a problem known as transcript spillover. This spillover is niche-dependent and, therefore, can confound downstream analyses that rely upon niche-specific differential expression. True spillover cannot be directly quantified in experimental ST data, so existing correction methods infer contamination indirectly from transcript positions or expression structure. Here we introduce SYNDRI, a post-segmentation correction method that frames spillover detection as a supervised, gene-level binary classification task, generating labeled training data with known spillover by adding transcripts from ST-informed simulated neighborhoods to spillover-free scRNA-seq profiles. SYNDRI uses a transformer trained on these synthetic examples to assign gene-level spillover probabilities from each cell's expression profile, cell identity, and local neighborhood, allowing the same gene to be identified as likely spillover in one cellular context but retained as endogenous expression in another. Across datasets from three spatial technologies, SYNDRI outperforms existing correction methods, as measured by cell-type separation, similarity to matched single-cell references, and preservation of within-cell-type heterogeneity. In a CosMx lung cancer dataset, SYNDRI is the only method that removes spillover-driven tumor and macrophage markers from differential expression between tumor and stromal fibroblasts while retaining native fibroblast markers. Together, these results establish SYNDRI as a general post-segmentation correction framework that can reduce spillover, preserve endogenous biological signal, and improve the reliability of imaging-based ST analysis pipelines.

1 Introduction

Imaging-based spatial transcriptomics (ST) measures the locations of individual RNA transcripts within intact tissue sections, enabling single-cell quantification of gene expression while preserving spatial context [1, 2]. Platforms such as Xenium [3], CosMx [2], and MERFISH/MERSCOPE [1] are now widely used to study tissue organization and how local microenvironments shape cellular identity and function [4]. Many of these studies rely on differential expression (DE) analysis to identify spatial transcriptional programs and characterize interactions between neighboring cell populations [5–8]. The reliability of these conclusions therefore depends on the accuracy of cell-level transcript counts.

However, cell-level counts are often confounded by transcript spillover, in which transcripts from one cell are assigned to a neighboring cell due to segmentation errors, optical crowding, or transcript diffusion [9]. Spillover distorts cell-type-specific expression profiles, introduces spurious DE genes [10], and can obscure true differences between cell populations [11, 12]. Because contaminating transcripts come from adjacent cells, the spillover a cell receives depends on its microenvironment. As a result, spillover is most problematic when comparing cells of the same type across different niches, which is central to many spatial analyses. For example, fibroblasts adjacent to tumor cells may appear to express tumor markers because of their location instead of their biology.

Several computational methods have recently been developed to correct spillover after segmentation. Because ground-truth spillover cannot be measured directly in ST data, these methods infer contamination without supervision, relying on assumptions about how spillover appears in the data. cellAdmix [11] and ovrlpy [13] use transcript coordinates to identify spillover, assuming that misassigned transcripts are spatially co-located. These approaches require transcript-level coordinates and are less sensitive to spillover that is intermixed with a cell's own transcripts, such as near the cell center. SPLIT [12] uses the deconvolution method RCTD [14] to estimate cell-type mixtures and then rescales counts based on a scRNA-seq reference, so its corrections depend on deconvolution accuracy and do not directly model each cell's spatial neighborhood. ResolVI [15] uses a latent-variable model to separate intrinsic expression from contamination and reconstructs corrected expression from its posterior instead of directly modifying observed counts. In each case, correction accuracy depends on how well the method's assumptions hold. There remains a need for a method that learns spillover patterns from labeled examples, accounts for each cell's local neighborhood, and removes contamination while preserving biological variation within cell types.

Here we present SYNDRI, a method that corrects transcript spillover by reframing it as a supervised learning problem. SYNDRI generates labeled training data by assigning spillover-free scRNA-seq profiles to the cell positions of an ST sample, producing a synthetic tissue with known expression. It then simulates spillover by transferring transcripts between neighboring cells in this synthetic tissue. A transformer trained on these data uses each cell's neighborhood to identify spillover specific to its local context. SYNDRI operates directly on cell-by-gene count matrices, does not require transcript coordinates or cell-type deconvolution, and only removes transcripts from the observed counts instead of resampling them. Across datasets spanning various species, tissues, disease contexts, and ST platforms we show that SYNDRI outperforms existing spillover correction methods in cell-type separation, similarity to reference expression profiles, and preservation of within-cell-type heterogeneity. We also show SYNDRI outperforms other methods directly using our simulated data with known spillover as a benchmark. In a CosMx non-small cell lung cancer dataset, SYNDRI was the only method that removed spillover-driven tumor and macrophage markers from the genes differentially expressed between fibroblasts in tumor and stromal regions. Source code for SYNDRI is available at https://github.com/gkrieg/syndri.

2 Methods

2.1 Overview of SYNDRI

SYNDRI corrects transcript spillover by learning from examples in which the spillover is known. Although spillover arises through several mechanisms, each adds transcripts from neighboring cells to a cell's measured expression profile (**Fig.** 1b). Endogenous and spillover transcripts contribute to the same observed counts and cannot be distinguished from those counts alone, so existing methods rely on fixed assumptions about how spillover appears. In contrast, scRNA-seq profiles are free of spatial spillover because cells are dissociated before sequencing, although they can contain ambient RNA [16, 17]. Spillover can therefore be simulated in these profiles with known ground truth. This moves assumptions about spillover from the model to the simulation, where they can be inspected and varied, and lets the model learn spillover patterns that are not explicitly specified. Simulation also provides an effectively unlimited supply of training examples for high-capacity architectures such as transformers.

To generate synthetic ST data with known spillover, SYNDRI-SIM assigns scRNA-seq profiles to the cell positions of the ST dataset by matching cell types, producing a synthetic tissue (**Fig.** 1a). It then adds transcripts from each query cell's spatial neighbors to its expression profile, so each simulated cell has known gene-level labels indicating which genes received spillover. We frame correction as gene-level binary classification: a transformer is trained on these examples to predict the probability that each expressed gene contains spillover, given the cell's expression profile, its cell type, and the cell types of its neighbors. The trained model is then applied to the real ST data, and genes are removed from each cell in descending probability order until a target fraction of transcripts has been removed, subject to a minimum probability threshold.

SYNDRI introduces several methodological advances for spillover correction in imaging-based ST. First, to our knowledge, it is the first method to treat spillover correction as a supervised learning problem. Second, it makes the removal decision directly, whereas existing methods reach it through intermediate tasks such as cell-type deconvolution or expression reconstruction, whose errors carry over into the correction. Third, its predictions depend on neighborhood context, so the same gene in cells of the same type can be removed in one niche and retained in another, addressing the niche dependence of spillover that produces spurious DE between microenvironments. Finally, SYNDRI-SIM provides ground truth for benchmarking spillover correction methods.

2.2 Synthetic Admixture Generation

The absence of experimentally measured ground-truth transcript spillover impedes direct supervised training on spatial transcriptomics data. We therefore construct synthetic training examples from scRNA-seq profiles, for which the origin of artificially introduced transcripts is known by construction. Each training example consists of a query cell, a set of neighboring source cells, a synthetically admixed expression profile, and binary gene-level labels indicating which genes received expression from the source cells.

2.2.1 Construction of Pseudo-Spatial Single-Cell Data

Because scRNA-seq measurements lack spatial coordinates, we construct a pseudo-spatial version of each single-cell sample before generating synthetic admixture, where we define admixture as the result of adding spillover to a cell's endogenous expression. For each dataset, we first select a representative ST slice whose cell-type adjacency distribution is most similar to the other slices in the dataset. To find this slice, we build a spatial KNN graph over the cells for each ST slice, and calculate the adjacency-enrichment profiles based on observed vs expected neighbor fractions computed from the cell type abundance for each directed cell-type pair. We use the log of the observed vs expected ratio for each pair as an adjacency-enrichment profile and select the slice with maximum mean Spearman correlation of its profile with every other slice.

The selected spatial slice and corresponding scRNA-seq reference are jointly embedded using Harmony [18]. Briefly, the two datasets are concatenated, library-size normalized, log-transformed, scaled,

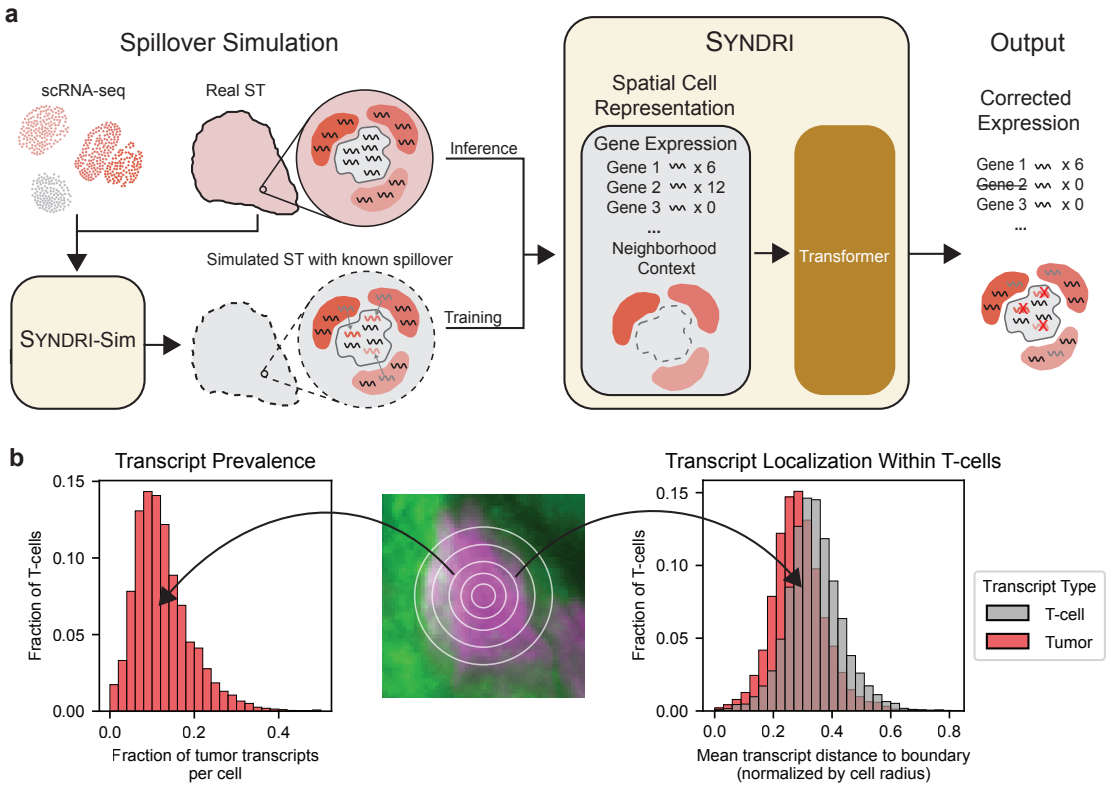


Figure 1: Overview of SYNDRI. a. SYNDRI frames spillover correction as supervised learning. During training, SYNDRI-SIM assigns spillover-free scRNA-seq profiles to the cell positions of a real ST sample and transfers transcripts between neighboring cells, producing simulated ST data with known spillover. SYNDRI takes each cell's gene expression and the identities of its neighboring cells and predicts which genes contain spillover. At inference, the trained model is applied to real ST data, and transcripts of genes predicted to be contaminated are removed from the observed counts. **b.** Histograms showing the distribution of the fraction of each T cell's transcripts that come from tumor-marker genes (left) and the distribution of mean distance of transcripts to the cell boundary (right). Tumor transcripts lie closer to the boundary than T cell transcripts, but are often found near the cell center.

and projected into a 30-dimensional PCA space. Harmony is then applied using data source (spatial or single-cell) as the batch covariate, yielding a shared representation for the spatial and single-cell profiles.

To transfer the observed tissue organization to the single-cell reference, we assign an scRNA-seq profile to each unique spatial-cell coordinate in the representative slice. For spatial cell r, let $\mathbf{h}_r^{\mathrm{ST}}$ denote its Harmony embedding and let $\mathbf{h}_j^{\mathrm{SC}}$ denote the embedding of single-cell reference cell j. We first identify the $M = 50$ nearest single-cell candidates to each spatial cell using Euclidean distance in the Harmony embedding,

$$d_{rj} = \left\| \mathbf{h}_r^{\mathrm{ST}} - \mathbf{h}_j^{\mathrm{SC}} \right\|_2 .$$

Candidate matches are restricted to those with the same cell-type annotation as the spatial cell and are converted to distance-based sampling weights,

$$w_{rj}^{(0)} \propto \exp\left(-\frac{d_{rj}^2}{\sigma^2} \right),$$

where σ is the median candidate-match distance across the mapping. To discourage a small number of reference cells from being repeatedly assigned throughout the pseudo-spatial tissue, these weights are further adjusted by a reuse penalty. If c_j denotes the frequency with which reference cell j occurs as the nearest

candidate across spatial positions, its sampling weight is proportional to

$$w_{rj} \propto w_{rj}^{(0)} \frac{1}{(c_j + \epsilon)^{\gamma}},$$

where γ controls the strength of the reuse penalty. The adjusted weights are normalized across the candidate cells for each spatial position, and one scRNA-seq cell is sampled from the resulting distribution. $\gamma = 9$ and a hard max reuse cap of 5000 per sample were used for all datasets.

The resulting pseudo-spatial dataset contains one sampled scRNA-seq expression profile for each unique spatial coordinate of the representative slice, and individual reference profiles may therefore be reused at multiple spatial positions. This procedure preserves the experimentally observed arrangement of cell types and local tissue neighborhoods while retaining scRNA-seq expression profiles as the substrate for synthetic admixture generation. Synthetic source cells – those from which synthetic spillover originates – are subsequently defined using physical proximity in this transferred coordinate system.

2.2.2 Sampling of Admixed Transcripts

For each query cell, we construct a candidate contaminating expression profile from its neighboring source cells. Let

$$\mathbf{s}_i = \sum_{j \in \boldsymbol{\mathcal{N}}_i} \mathbf{x}_j$$

denote the aggregate expression of the source-cell neighborhood.

To make synthetic contamination identifiable at the gene level, candidate admixed genes are restricted to genes not already detected in the query cell $\boldsymbol{\mathcal{G}}_i^{\text{admix}} = \{g : x_{ig} = 0 \text{ and } s_{ig} > 0\}$. Thus, any nonzero expression introduced at an eligible gene can be attributed unambiguously to synthetic admixture.

For each training example, a target admixture level τ is sampled uniformly from a predefined range ($[0.7, 0.85]$ for our models), corresponding to the proportion of expression in the admixed profile attributed to admixed expression.

The target number of contaminating transcripts is therefore

$$T_i^{\text{admix}} = \frac{\tau}{1 - \tau} \left\| \mathbf{x}_i \right\|_1 .$$

Eligible genes from source cells are randomly permuted, and whole gene counts are added sequentially until the cumulative number of available source-cell transcripts reaches the target admixture level.

2.2.3 Gene-Level Admixture Labels

Because the synthetic admixture procedure explicitly records which genes receive contaminating transcripts, each training example is paired with a binary gene-level target. For query cell i and gene g, $y_{ig} = \mathbb{I}[a_{ig} > 0]$, where a_{ig} is the total number of synthetic contaminating transcripts added for gene g. Thus, $y_{ig} = 1$ indicates that the observed expression of gene g contains synthetically introduced signal, while $y_{ig} = 0$ indicates that no synthetic admixture was added for that gene. These labels provide direct supervision for the gene-level admixture prediction task used to train SYNDRI.

2.3 SYNDRI Architecture

Admixture cannot always be identified from the expression of an individual gene in isolation. Whether expression is plausible depends on the identity of the query cell, its complete expression profile, and the types of cells present in its local spatial environment. We use a transformer architecture previously developed by our group that jointly represents gene expression and cellular context [19].

Table 1: Summary of spatial transcriptomics datasets and matched single-cell references used in this study.

Abbrev.	Dataset	Technology	Spatial cells	Genes	Reference cells	Slices / donors
BM	Whole mouse brain	MERFISH	215,278	1,122	79,989	3
NC	NSCLC	CosMx	771,236	956	40,362	8
BX	Breast cancer	Xenium	1,673,926	277	10,658	19

2.3.1 Gene Identity and Expression Tokenization

For each query cell, the input contains one token for each gene with nonzero observed expression. Gene tokens are ordered by decreasing expression, as introduced in Geneformer [20]. Each token combines a learned embedding of the gene identity with a learned linear projection of its expression count, where the counts are neither normalized nor log-transformed.

2.3.2 Cell-Type and Neighborhood Tokenization

The gene tokens are preceded by metadata tokens describing cellular context. For each query cell, we include one learned token for the cell type of each of its five spatial neighbors, followed by a learned token for the query cell's annotated cell type. Thus, the input represents the query expression profile together with the types of the cells in its local neighborhood and the query cell's own type. Finally, we prepend a `[CLS]` token to the complete sequence.

2.3.3 Transformer Encoder and Prediction Head

The input embeddings are augmented with learned, BERT-style positional embeddings and processed by a four-layer transformer encoder with eight self-attention heads and a hidden dimension of 128. Each layer uses a 512-dimensional feed-forward sublayer with GELU activation; dropout is 0.1.

For admixture prediction, a shared linear head is applied to each contextualized gene-token embedding. The `[CLS]` and six cell-type prefix-token embeddings are excluded from gene-level prediction. For query cell i and gene g, let $\mathbf{h}_{ig} \in \mathbb{R}^{128}$ denote the contextualized gene-token representation. The head maps each gene embedding to a scalar admixture logit, $z_{ig} = \mathbf{w}^{\mathsf{T}}\mathbf{h}_{ig} + b$, from which the gene-level admixture probability is obtained as $p_{ig} = \sigma(z_{ig})$.

Although predictions are produced separately for each gene token, each $\mathbf{h}_{ig}$ incorporates information from the complete input sequence through transformer self-attention [21]. Predictions can therefore depend on the query expression profile, its cell-type annotation, and the cell types in its spatial neighborhood.

2.4 Admixture Correction

After inferring gene-level admixture probabilities with a trained SYNDRI model, we construct the corrected count matrix using a probability-ranked removal strategy. For each cell, genes are ranked by predicted admixture probability, and the entire observed count for a gene is removed in descending probability order until a predefined target proportion of transcripts, ρ, has been removed. We fixed ρ at 0.5 for each dataset. To avoid removing genes with weak evidence of admixture, we also impose a minimum probability of 0.5, so genes with predicted admixture probability below 0.5 are never removed.

3 Results

3.1 SYNDRI improves cell-type separation while preserving reference similarity and within-cell-type heterogeneity

We benchmarked SYNDRI against SPLIT [12], ResolVI [15], ovrlpy [13], and cellAdmix [11], along with the uncorrected data, on whole mouse brain MERFISH (BM), non-small cell lung cancer CosMx (NC), and breast cancer Xenium (BX) datasets (**Tab.** 1). ovrlpy and cellAdmix require transcript-level coordinates and could therefore be evaluated only on NC and BX (gray entries in **Fig.** 2b). Because spillover

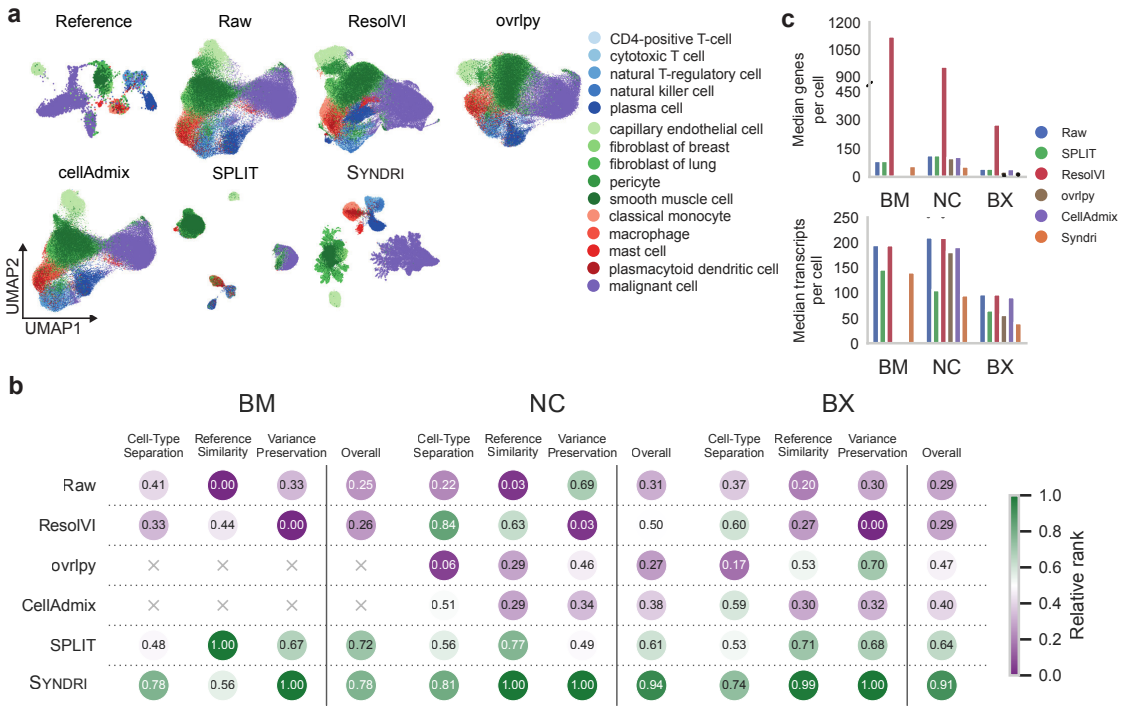


Figure 2: Benchmarking of spillover correction methods. **a.** UMAP [22] embeddings of the breast cancer Xenium (BX) scRNA-seq reference and of a single spatial slice, uncorrected (Raw) and after correction by each method, colored by cell type. **b.** Relative rank of correction methods for three datasets under three criteria: cell-type separation, cosine similarity to the matched single-cell reference (Reference Similarity), and Spearman correlation of gene-wise variance with the reference (Variance Preservation). Ranks were computed per slice and averaged across slices within each dataset, and Overall is the unweighted mean of the three criteria. Gray crosses mark evaluations that could not be performed. **c.** Median number of detected genes (top) and transcripts (bottom) per cell for each method in each dataset.

correction can fail in distinct ways, we scored each method on three criteria (**Fig.** 2): cell-type separation, which should improve as cross-cell-type contamination is removed; reference similarity to a matched, comparatively uncontaminated single-cell reference; and preservation of within-cell-type heterogeneity, which guards against over-smoothing. We ranked methods on each metric per slice, averaged ranks across slices within each dataset, and weighted the three criteria equally in the aggregate score (Appendix A).

Relative to the uncorrected data, both SPLIT and SYNDRI visibly improved cell-type separation in BX (**Fig.** 2a). However, SPLIT retained overlap between several cell populations, including fibroblast-associated structure that was discordant with the matched single-cell reference annotations. This overlap likely reflects SPLIT's reliance on RCTD deconvolution: where RCTD assigns ambiguous or incorrect cell-type mixtures to a cell, SPLIT's rescaling inherits those errors (**Fig.** S1). SYNDRI, which does not depend on deconvolution, produced clearer separation while maintaining overall correspondence with the matched single-cell reference. In contrast, ResolVI, ovrlpy, and cellAdmix showed little change in cell-type separation relative to the uncorrected data.

Quantitatively, SYNDRI achieved the highest aggregate score in all three datasets, with the largest margins over competing methods in the two cancer datasets, NC and BX (rankings in **Fig.** 2b; metrics in **Fig.** S2). It ranked first or second for cell-type separation in every dataset and first for heterogeneity preservation in every dataset. All correction methods improved reference similarity relative to the uncorrected data, consistent with spillover shifting spatial expression profiles away from the matched single-cell reference. SYNDRI achieved the highest reference similarity in NC and BX, whereas SPLIT ranked first in BM.

ResolVI displayed a different performance profile. Although it frequently achieved high silhouette scores, it performed substantially worse on ARI and NMI [23] and consistently showed the lowest het-

erogeneity preservation (**Fig.** S2c). This pattern follows from its generative design: ResolVI reconstructs expression by sampling from a learned posterior, so many originally zero entries receive small fractional values, which increases the number of detected genes per cell while leaving total transcript abundance similar (**Fig.** 2c). After library-size normalization and log transformation, these small values can contribute appreciably to downstream representations, inflating silhouette scores while degrading within-cell-type structure. This discrepancy illustrates why cell-type separation alone is insufficient to evaluate correction quality: a method can produce compact, well-separated embeddings while substantially altering within-cell-type expression structure.

Overall, competing methods each traded one criterion against another: SPLIT inherited errors from its deconvolution step, ResolVI sacrificed within-cell-type structure for separation, and ovrlpy and cellAdmix yielded limited gains in separation. By learning spillover patterns from labeled synthetic examples and only removing counts, SYNDRI was the only method to perform consistently well on all three criteria across species, tissues, and platforms.

3.2 SYNDRI outperforms existing methods on simulated data with known spillover

Because ground-truth spillover cannot be measured in ST data, the benchmarks above rely on indirect measures. SYNDRI-SIM builds simulated ST data from spillover-free scRNA-seq profiles and therefore records the true spillover of every gene in every cell, so it can also serve as a ground-truth benchmark. We first asked how closely the simulated data resemble real ST data in the NC dataset. Adding spillover with SYNDRI-SIM reduced the RBF MMD^2 [24] between cell-type pseudobulks of the single-cell data and the matched spatial counts used to define the cell layout by around 30% across all three replicates (**Fig.** 3a). Because SYNDRI-SIM assigns single-cell profiles to the cell positions of a real sample, the simulated data matched the cell-type adjacencies of the real slice closely (**Fig.** 3b), including the enrichment of each cell type among its own neighbors and the depletion of malignant cells next to immune cells. We also compared the co-detection of marker genes from different cell types, which should be rare in uncontaminated cells (**Fig.** 3c). In the scRNA-seq reference, such co-detection was largely limited to related immune populations, whereas the real ST slice showed much broader co-detection, particularly involving fibroblast, epithelial, and malignant markers. SYNDRI-SIM introduced co-detection in these same groups of markers, though it did not reproduce every pattern in the ST slice. This gap may arise because real spillover also comes from cells above and below the imaged plane, whereas SYNDRI-SIM transfers transcripts only between neighboring cells in two dimensions, so the simulated and real data can match in adjacency yet differ in co-detection.

We used SYNDRI-SIM to compare SYNDRI with SPLIT and ResolVI. ovrlpy and cellAdmix were excluded because they require transcript coordinates, which SYNDRI-SIM does not generate. The NC single-cell reference contains data from 7 donors. We simulated training data from 5 donors for SYNDRI, used 1 for validation, and evaluated all methods on simulated data from the remaining 1. SPLIT used the combined training and validation donors as its reference, and ResolVI was fit on the test slice.

SYNDRI-SIM adds spillover only to genes that the query cell does not endogenously express. For each gene in each cell, we therefore scored removal as $s = 1 - c/a$, where c and a are the corrected and admixed counts, so that $s \leq 1$ and higher values indicate stronger removal. We evaluated these scores in two ways: as real values, ranking genes by s to compute the AUPRC, and as binary calls. For SYNDRI, the real-valued score was its predicted admixture probability and the binary call was its post-threshold removal. For SPLIT and ResolVI, we chose the binarization threshold that maximized F1-score, which requires ground truth that is unavailable in practice.

SYNDRI outperformed both methods under both scoring schemes (**Fig.** 3d). With real-valued scores, SYNDRI reached an AUPRC of 0.89, compared with 0.53 for ResolVI and 0.54 for SPLIT. With binarized scores, SYNDRI reached 0.80, compared with 0.58 and 0.57, even though its threshold was set by its standard procedure instead of being tuned on the ground truth.

This benchmark has limitations. SYNDRI is trained on data from the same simulator, although on dif-

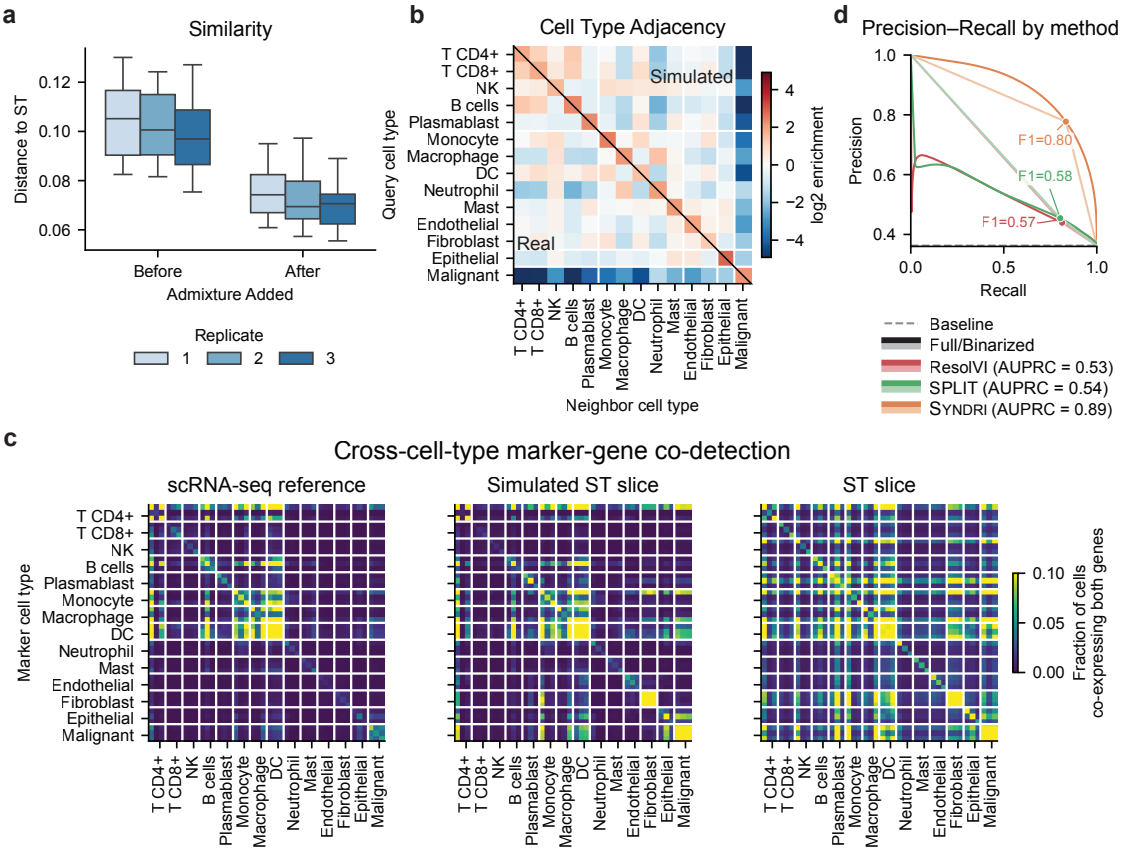


Figure 3: Evaluation of spillover correction on simulated spatial data with known ground truth. a. Distance (RBF MMD^2) between cell-type pseudobulks of the mapped scRNA-seq data (Before) or the simulated ST data (After) and those of the matched real ST data, for three ST slices. **b.** Cell-type adjacency in a single slice of real ST data (lower triangle) and simulated ST data (upper triangle), shown as the $\log_2$ ratio of the observed frequency of each neighbor cell type among the neighbors of each query cell type to the frequency expected from overall cell-type abundance. **c.** Co-detection of the top three marker genes from different cell types, shown as the fraction of cells expressing both genes of each pair, in a single slice for the scRNA-seq reference (left), simulated ST data (middle), and real ST data (right). **d.** Precision–recall curves for SYNDRI, SPLIT, and ResolVI at identifying simulated spillover across the three test slices. Dark curves show real-valued removal scores $1 - c/a$, where c and a are the corrected and admixed counts, with AUPRC given in the legend. Light curves show binarized calls, with points marking the threshold that maximizes the F1-score for SPLIT and ResolVI and the corresponding F1-score. For SYNDRI, real-valued scores were predicted admixture probabilities. The dashed line indicates a random baseline.

ferent donors, so it is evaluated on spillover of the type it was trained to remove. Because spillover is added only to genes not endogenously expressed by the query cell, the task favors binary detection, which does not match how ResolVI and SPLIT adjust counts; we therefore report both real-valued and binarized scores. SYNDRI-SIM also does not fully reproduce the marker co-expression of real ST data. Nevertheless, the simulation captures key co-expression patterns of real ST data and gives a setting with known spillover, in which SYNDRI substantially outperformed existing methods.

3.3 SYNDRI enables more reliable downstream biological analyses

Spatial differential expression (DE) analysis underlies many downstream analyses, so we tested whether spillover correction improves its conclusions in the NC dataset (**Fig.** 4). Following the analysis introduced in cellAdmix [11], we compared fibroblasts located in tumor regions with fibroblasts located in stromal regions. From markers identified with pagoda2 [25] (adjusted $p < 0.01$, log-fold change $> \log(1.5)$), we selected malignant and macrophage markers that are nearly absent from fibroblasts in the scRNA-seq

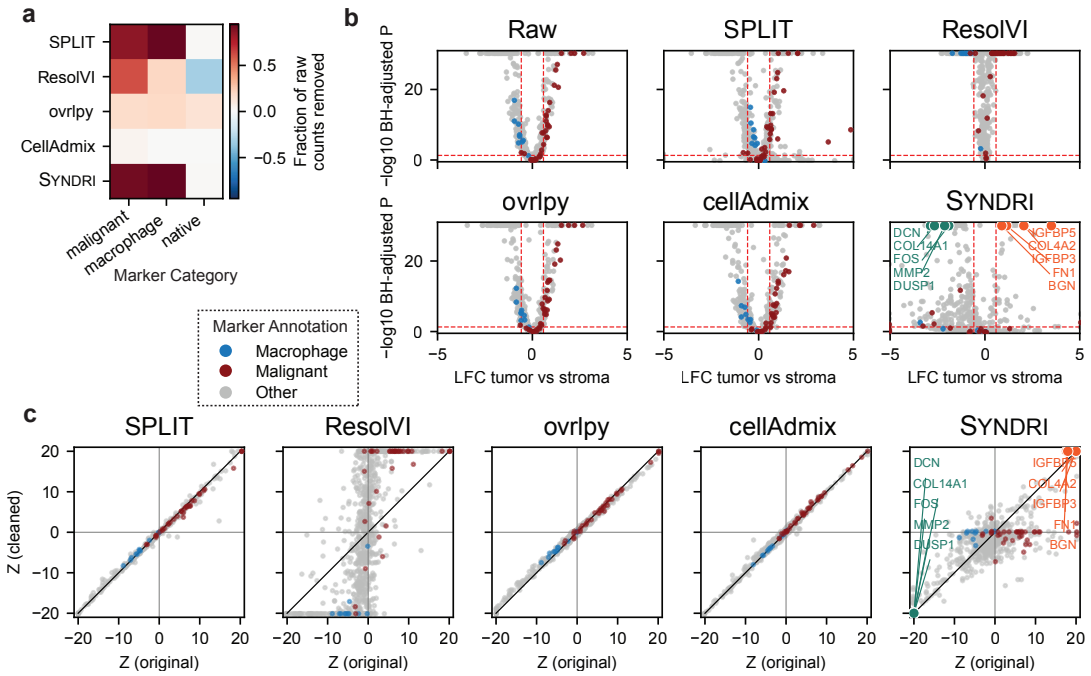


Figure 4: SYNDRI removes spillover-driven tumor and macrophage markers from differential expression in CosMx NSCLC fibroblasts. a. Fraction of raw transcript counts removed by each correction method for malignant, macrophage, and native fibroblast marker sets in annotated fibroblasts. **b**. Volcano plots of differential expression between fibroblasts in tumor and stromal regions, pooled across three replicates from one donor, before correction (Raw) and after correction by each method. Discussed DE genes for stroma and tumor are highlighted in green and orange, respectively. **c**. Comparison of gene-wise Wilcoxon Z statistics for tumor-versus-stroma differential expression before and after correction. The diagonal denotes unchanged differential signal and movement toward $Z = 0$ indicates attenuation of the original differential expression. The same genes from **b** are highlighted for SYNDRI.

reference and treated them as candidate foreign signals. A separate set of native fibroblast markers served to assess preservation of endogenous expression.

We first quantified how strongly each correction method removed transcripts belonging to these three marker sets by calculating the fraction of transcripts present in the uncorrected data that were removed after correction (**Fig.** 4a). SYNDRI removed nearly all malignant-marker transcripts and a similarly large fraction of macrophage-marker transcripts, while removing very little native fibroblast-marker expression. SPLIT also removed substantial malignant and macrophage signal, but less completely than SYNDRI. In contrast, ovrlpy and cellAdmix removed little of the malignant-marker signal, and ResolVI produced reconstructed expression values instead of a directly corrected observed count matrix, complicating interpretation of transcript removal. Although cellAdmix was originally evaluated using this fibroblast differential-expression task, our rerun using the recommended settings ("cell-bridging" and factors $= 1.5\times$ number of cell types) did not reproduce the results of the original study, which showed better malignant and macrophage signal removal.

These differences changed the DE conclusions. In the uncorrected data, many significant DEGs were malignant or macrophage markers, reflecting neighboring tumor and immune cells rather than the fibroblast program (**Fig.** 4b). SYNDRI was the only method after which nearly all of these markers became non-significant; they remained among the DEGs after SPLIT, ovrlpy, and cellAdmix. ResolVI produced a substantially different DE landscape, likely because its posterior reconstruction assigns small fractional values to many genes, including genes not detected in the original data. Comparing gene-wise Wilcoxon Z statistics before and after correction showed the same pattern (**Fig.** 4c): for SPLIT, ovrlpy, and cellAdmix, most genes, including most malignant and macrophage markers, stayed near the diagonal, whereas SYN-

DRI moved nearly all malignant and macrophage markers toward $Z = 0$. Many non-marker genes retained substantial DE signal under SYNDRI.

The strongest DEGs remaining after SYNDRI correction were associated with fibroblast state and matrix organization. Tumor-region fibroblasts were enriched for *IGFBP5*, *COL4A2*, *IGFBP3*, *FN1*, and *BGN*, and stromal fibroblasts for *DCN*, *COL14A1*, *FOS*, *MMP2*, and *DUSP1*. These are consistent with the heterogeneous fibroblast states described across tissues and in cancer [26], including adventitial, *COL14A1*-associated matrix-fibroblast programs in human NSCLC [27] and murine lung [28]. Removing malignant and macrophage signal therefore did not collapse biological variation among fibroblasts.

Together, these results show that spillover distorts spatial DE analysis and that methods differ greatly in their ability to remove it. SYNDRI removed malignant and macrophage signal from fibroblasts while preserving native fibroblast markers and plausible fibroblast-state differences, improving the reliability of downstream analyses such as marker discovery and tumor–stroma comparisons.

We also evaluated how SYNDRI responds to neighborhood context by holding each query cell's expression fixed and changing only its neighbors' cell types, and found that SYNDRI adjusts its predictions in a cell-class- and gene-specific manner (Appendix B, **Fig.** S3).

4 Discussion

We developed SYNDRI, a framework that corrects transcript spillover in imaging-based spatial transcriptomics by casting it as a supervised learning problem. SYNDRI generates labeled training data by assigning spillover-free scRNA-seq profiles to the cell positions of an ST sample and simulating contamination between neighboring cells, so that the true expression of every cell is known. A transformer trained on these data uses each cell's expression, cell type, and neighborhood to predict an admixture probability for every observed gene, and SYNDRI removes genes predicted to be contaminated. This design lets SYNDRI operate directly on cell-by-gene count matrices, without transcript coordinates or deconvolution.

Across MERFISH, CosMx, and Xenium datasets, SYNDRI outperformed existing methods in cell-type separation, reference similarity, and heterogeneity preservation, improving cell-type separation without sacrificing within-type variation. In NSCLC, it removed spillover-driven tumor and macrophage markers from the DEGs between fibroblasts in tumor and stromal regions while retaining DEGs consistent with known fibroblast states. Controlled context-remapping experiments further showed that SYNDRI's predictions change with neighborhood composition in a cell-class- and gene-specific manner.

Looking ahead, SYNDRI could be extended in several directions. Our synthetic spillover only transfers transcripts between neighboring cells. Training could instead model more realistic mechanisms, such as gene-specific diffusion, segmentation geometry, or transcript density. The neighborhood representation could also include continuous distances, morphology, or graph-based spatial context in place of discrete cell-type tokens. In addition, uncertainty-aware correction could replace the current fixed removal criterion when endogenous and contaminating expression are difficult to distinguish. More broadly, synthetic supervision from clean molecular profiles could apply to other settings where ground truth is unavailable.

In general, SYNDRI provides a flexible tool for removing spillover while preserving biological variation, supporting more reliable analysis of cellular organization and gene-expression programs across spatial technologies and tissues.

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